

Program2

Develop a Multi-Stage Dockerfile for container Orchestration

step1: project structure:

```
|--Program2
  |--src
  |--dist
    |--index.js
  |--Dockerfile
  |--build
  |--package.json
  |--package-lock.json
  |--node_modules
```

step2: create a Dockerfile

step3: create index.js inside index.js

step4: create package.json

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ cd program2
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ nano Dockerfile
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ cd dist
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2/dist$ nano index.js
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2/dist$ cd ..
```

run >> npm init -y

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ npm init -y
Wrote to /home/rv24mc053_kavya/program2/package.json:

{
  "name": "program2",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  },
  "devDependencies": {},
  "description": ""
}

rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$
```

step5: >>npm install express

>>npm install

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ npm install express

up to date, audited 69 packages in 1s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ npm install

up to date, audited 69 packages in 1s

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found 0 vulnerabilities
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$
```

step 6 : build the dockerfile

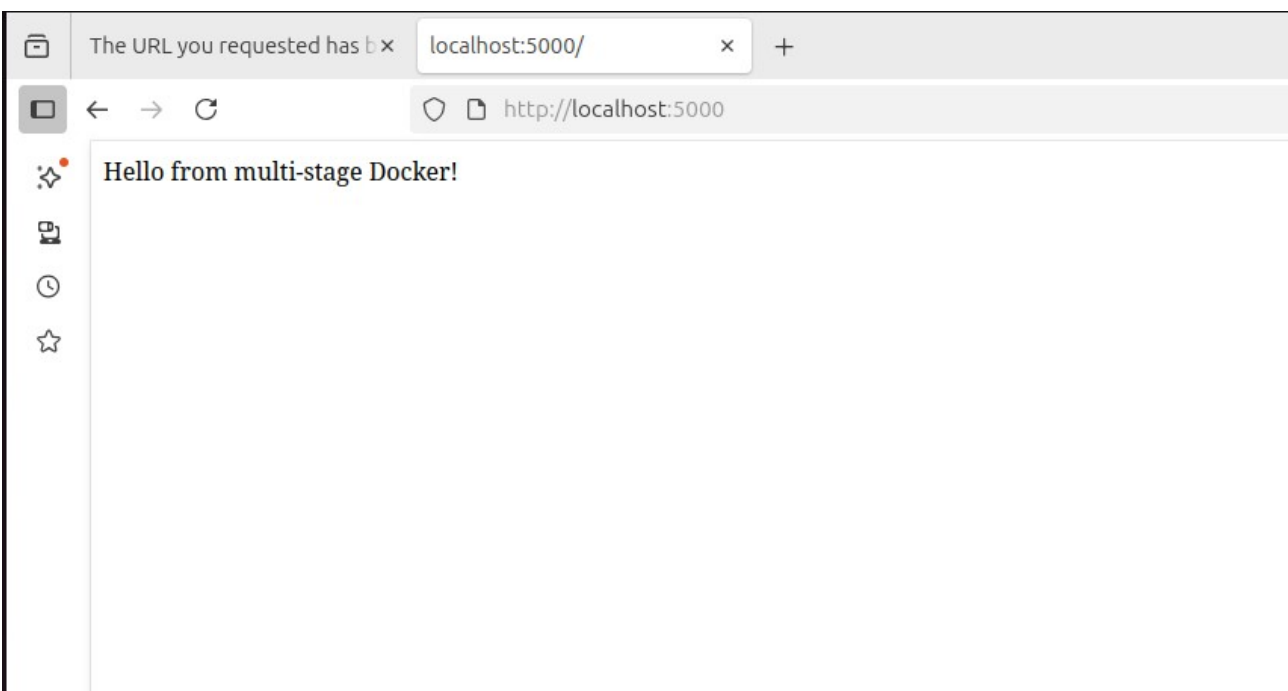
>> docker build -t program2 .

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ docker build -t program2 .
[+] Building 2.7s (13/13) FINISHED
=> [internal] load build definition from Dockerfile
=> == transferring dockerfile: 560B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> == transferring context: 2B
=> [builder 1/5] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa
=> [internal] load build context
=> == transferring context: 44.38kB
=> CACHED [builder 2/5] WORKDIR /app
=> CACHED [builder 3/5] COPY package.json package-lock.json ./
=> CACHED [builder 4/5] RUN npm install
=> CACHED [builder 5/5] COPY . .
=> CACHED [stage-1 3/5] COPY --from=builder /app/package.json ./
=> CACHED [stage-1 4/5] COPY --from=builder /app/package-lock.json ./
=> CACHED [stage-1 5/5] COPY --from=builder /app ./
=> exporting to image
=> == exporting layers
=> == writing image sha256:c61f656a07c84e605d2d32fe02c5ca7284954e6aa66acea3f0ad5d9613c6f87b
=> == naming to docker.io/library/program2
```

Step7: expose to the port and check the output in browser

>> docker run -p 5000:5000 program2

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~/program2$ docker run -p 5000:5000 my-node-app
Server running on port 5000
```



>>nano Dockerfile

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx: ~/program3
GNU nano 7.2 Dockerfile
# =====
# Stage 1: Builder Stage
# =====
FROM node:20-alpine AS builder

WORKDIR /app

# Copy package files and install dependencies
COPY package.json package-lock.json ./
RUN npm install

# Copy the rest of the app (including dist/)
COPY . .

# =====
# Stage 2: Production Stage
# =====
FROM node:20-alpine

WORKDIR /app

# Copy package files and installed node_modules from builder
COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/node_modules ./node_modules

# Copy app source files (including dist/)
COPY --from=builder /app ./

# Expose port
EXPOSE 5000

# Run the app
CMD ["node", "dist/index.js"]
```

>>nano dist/index.js

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx: ~/program3
GNU nano 7.2 index.js
// index.js
const express = require('express');
const app = express();

// Use PORT from environment variable, or default to 5000
const PORT = process.env.PORT || 5000;

// Define a simple route
app.get('/', (req, res) => {
  res.send('Hello from multi-stage Docker!');
});

// Listen on all network interfaces
app.listen(PORT, '0.0.0.0', () => {
  console.log(`Server running on port ${PORT}`);
});
```

